

① cloud computing is a delivery of computing service such as servers, storage, data bases networking, software and analytics over the internet on a pay as you basis. Instead of purchasing & maintaining physical users.

Characteristics of cloud computing

1. on-demand self service

→ users can create, configure, and use computing resources whenever required without contacting the service provider

2. Broad network Access

→ cloud services are available over the internet & can be accessed using laptops, desktops, smartphones

② cloud computing evolved through several stages of technical developments each stage improved the efficiency, accessibility and scalability of computing resources and leading to the modern cloud

1. Hardware Evolution

→ In the early days, organisations used large mainframe computers for processing data

2. Network Evolution

→ computers were connected using local Area Network & wide Area Networks

→ This laid the foundation for distributed computing

3. Internet Evolution

→ The growth of the internet allowed computers across the world to communicate

③ server virtualisation is the process of creating multiple virtual machines on a single physical using a software on a software called a hypervisor.

Role of server virtualisation

→ Increases hardware utilization by running multiple VMs on one server ✓

→ Reduces hardware, electricity and maintenance costs

virtualisation

cloud computing

* Technology that creates virtual machines

* Service model that delivers computing resources

* Focuses on efficient hardware utilisation

* Focuses on providing scalable and on demand

* Can exist without cloud computing

* Usually and on demand services

④ A data center is a secure facility that contains servers, storage devices, networking equipment, power supply and cooling systems used to provide cloud services

Role of data centers

* Stores & manages large amounts of customer data

* Hosts cloud applications & virtual machines

* Provides continuous service with backup power and redundant systems

- * supports disaster recovery through data backup replication
- * Enables scalability by adding servers to increasing demand
- * Google cloud data centers provide reliable cloud service to users

⑤ service model

1) IaaS

Infrastructure as a service

Ex:- Amazon EC2

2) PaaS

Platform as a service

Ex:- Google APP Engine

3) SaaS

Software as a service

Ex:- gmail or microsoft 365

4) XaaS

Anything as a service

Ex:- microsoft Azure